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Illumination schemes for coded coherent diffraction imaging: A comprehensive comparison

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ABSTRACT

Coded coherent diffraction imaging (CDI) introduces physical constraints to the conventional CDI by wavefront modulation, providing data redundancy for successful phase retrieval. In coded CDI, an object is modulated by several pre-defined illumination patterns that are either real or complex. The iterative phase retrieval algorithms are typically adopted to reconstruct the object wavefront. We note that different illumination schemes might affect imaging efficiency and achievable reconstruction quality. However, there has been no comprehensive review discussing respective advantages, which is important for further applications and development of coded CDI. In this study, we comprehensively investigate six representative illumination pattern structures in a unified framework. The illumination patterns adopted for both amplitude and phase modulation include binarized random, binarized blue noise, complementary, incremental, gray-scale random, and gray-scale blue noise patterns. To provide a fair platform from an algorithmic point of view, the reconstruction algorithms are based on the ptychographic iterative engine (PIE) framework with both serial and parallel structures. The comparison from the aspects of imaging quality, imaging efficiency, and robustness to noise not only reveals the connections and differences among these illumination schemes, but also shows the advantages, limitation, and suitable applications of their own.

1. Introduction

As existing optoelectronic detectors only yield amplitude measurements, phase problem [1] is a fundamental limitation inherent to coherent imaging systems for optical [2–4], crystallographic [5], and astronomical applications [6]. Coherent diffraction imaging (CDI) [7] has emerged as a promising non-interferometric technique that computationally retrieves phase information from the recorded intensity-only diffraction patterns using sophisticated phase retrieval algorithms [1]. However, the current single-shot CDI technique [8,9] is still limited to isolated specimens, or requires the use of well-characterized and localized illumination [10]. Besides, it is always accompanied with problems of ambiguous solutions [11,12] and convergence stagnation [13], which are particularly critical when the intensity information contains measurement noise.

Multi-image phase retrieval techniques involving different types of diversities in data collection are effective strategies to overcome these restrictions. Robust phase retrieval is typically achieved by introducing defocus multi-height diversity [14–17], multi-wavelength diversity [18–20], transverse translational diversity in the spatial [21–23] and the Fourier domain [24], among others. However, all these strategies require extra controllable mechanical devices or wavelength calibration and dispersion compensation processes, resulting in a significant increase in the complexity of the whole system [25]. Also, the common mechanical scanning or wavelength scanning makes the data acquisition process relatively time-consuming, thus leading to high requirements for the stability of the imaging system [26].

In the past few years, pre-defined wave modulation, which introduces physical constraints to the conventional CDI without mechanical scanning, has received great attention in the phase retrieval community. Inserting a coded mask between the sample and the detector [27–29] removes the twin image and spatial shift ambiguities of conventional CDI. The requirements for object size and support accuracy can also be alleviated. However, the technique is still limited to isolated or sparse objects. Multi-image coded CDI modulates an object by several pre-defined illumination patterns that are either real [30–36] or complex [37–40]. The patterns are typically implemented using a digital micromirror device (DMD) or a spatial light modulator (SLM), thanks to their advantages in real time and the programmable diversity of the wavefronts. Specifically, a DMD is a micro-electromechanical

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system with an array of switchable micromirrors [41]. Having two stable states (i.e., mirror surface at +12 deg or "on" state and at −12 deg or "off" state), a DMD modulates the amplitude distribution of a light field in a pixelated manner by switching the "on" and "off" state of each micromirror. The binary amplitude modulation can be performed at a refreshing rate of up to 30 kHz [42]. Gray-scale amplitude modulation can also be implemented by a DMD using temporal pulse width modulation, however, the refreshing rate will be dramatically reduced to less than 100 Hz [43]. A phase-type SLM, on the other hand, perturbates the phase of a wavefront. Currently available commercially SLMs can usually perform binarized and gray-scale phase modulation. However, they are typically based on liquid crystals on silicon (LCoS), making them still limited in refreshing rate to sub kHz due to the viscoelasticity of the liquid crystal material [44]. The advantage of phase modulation, compared to amplitude modulation, is the lossless energy transmission of an incident wavefront through the modulator, which helps reduce the number of measurements for comparable reconstruction quality [45]. Binary amplitude modulation, on the contrary, has attracted great attention due to its high refreshing rate and the advantage of dispensing with the exact modulation function of the SLM.

The structure and the number of illumination patterns are crucial inasmuch they determine the acquisition time and achievable phase retrieval quality. It was shown in Ref. [46] that the spatial structures of binarized amplitude patterns have significant influence on the reconstruction quality and the required number of projections. For phase modulator design, two considerations were discussed in Ref. [47], including facilitating the data acquisition and improving the convergence of the algorithm. Then, when the illumination patterns and corresponding diffraction measurements are known, the iterative phase retrieval algorithms can be adopted to reconstruct the complex-field transmission of the target, regardless of the illumination strategy used. The algorithms can be computed in a serial or parallel manner [48]. The former requires that the output of the previous iteration be used as the input of the next estimation and update operation is accomplished by altering the illumination pattern, while the latter means that multiple measured images participate simultaneously in calculation and have no influence on each other. We note that different illumination schemes and reconstruction algorithms might affect imaging efficiency and achievable reconstruction quality. Also, each scheme or algorithm has its own advantages, limitation, and suitable applications. Therefore, it is necessary to take the representative illumination schemes and reconstruction algorithms into comparison, and investigate their performance from various aspects under the same framework, to offer researchers an easy choice for their experiments.

In this study, we comprehensively investigate six representative illumination pattern structures in a unified framework. The illumination patterns adopted for both amplitude and phase modulation include binarized random, binarized blue noise, complementary, incremental, gray-scale random, and gray-scale blue noise patterns. To provide a fair platform from an algorithmic point of view, the reconstruction algorithms are based on the ptychographic iterative engine (PIE) framework [21] with both serial and parallel structures. We introduce in detail how the patterns are designed and how the algorithms are implemented. The comparison from the aspects of imaging quality, imaging efficiency, and robustness to noise not only reveals the connections and differences among these illumination schemes, but also shows the advantages and disadvantages of their own.

The remainder of this paper is organized as follows. Modeling of the coded CDI system and description of different illumination schemes and reconstruction algorithms are presented in Section 2. The comparisons under different experimental settings are conducted in Section 3. Finally, we conclude this paper in Section 4.

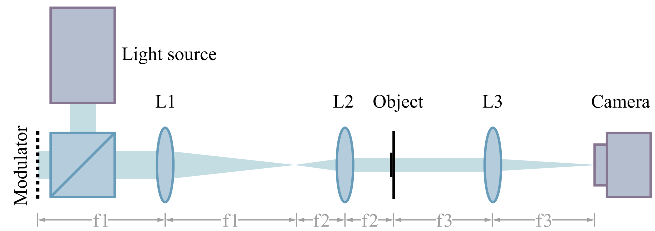

Fig. 1. Schematic diagram of coded CDI system. The coherent light source is first modulated by a modulator. A series of illumination patterns are sequentially projected onto the object using a 4f lens configuration (L1 and L2). A camera is placed at the Fourier plane of the object to capture corresponding diffraction images. The measurements are input into the PIE-based algorithms to recover both the object's amplitude and phase. L1–L3, lens; f1–f3, focal length.

2. Principle

2.1. Image formation

A schematic of our coded CDI system is shown in Fig. 1. First, the collimated coherent light source is sequentially modulated by a group of pre-generated illumination patterns with a modulator. The illumination patterns can be either amplitude or phase ones. The object with complex transmittance is then illuminated by the masked light using a 4f lens configuration. This provides sufficient structural information to condition the inverse problem solved by the reconstruction algorithm. The light then goes through a lens and a camera placed on the back focal plane of the lens collects the resulting diffraction images for further processing. Mathematically, the measurement formation can be described as

$$I_k = \left|\mathcal{F}\left(P_k \odot O\right)\right|^2 \quad (k = 1, 2, \ldots, K), \tag{1}$$

where $I_k \in \mathbb{R}^{m\times n}$ is the kth recorded intensity image, P_k represents the kth illumination pattern ($P_k \in \mathbb{R}^{m\times n}$ for amplitude modulation and $P_k \in \mathbb{C}^{m\times n}$ for phase modulation), $O \in \mathbb{C}^{m\times n}$ denotes the complex-valued object, $\odot$ stands for entry-wise multiplication, and $\mathcal{F}$ is the two-dimensional Fourier transform. Specifically, the illumination pattern P_k is projected onto the object O, which produces the object exit wave $P_k \odot O$. Then, the exit wave goes through a lens. The process can be approximated as two-dimensional Fourier transform. The wavefront at the camera's plane is therefore $\mathcal{F}\left(P_k \odot O\right)$. Finally, the camera records the intensity of the wavefront, yielding the recorded intensity image $I_k = \left|\mathcal{F}\left(P_k \odot O\right)\right|^2$.

2.2. Illumination schemes for coded CDI

In this subsection, we describe in detail the types of illumination patterns used for comparison and how they are generated. Six representative pattern structures are selected, including binarized and gray-scale ones, spatially inhomogeneous and homogeneous ones, and ones with different transmittances. Illustrations of the included six structures are shown in Fig. 2. The illumination patterns are adopted for both amplitude and phase modulation.

2.2.1. Binarized random patterns

To achieve high-quality phase retrieval, the importance of the randomness of the patterns is emphasized in Ref. [49,50]. Binarized random patterns are widely used in coded imaging systems [33,34,46], since the randomness is guaranteed. Fig. 2(a) illustrates a binarized random pattern with $\frac{1}{2}$ transmittance, where the white elements indicate one-value entries and the black elements are zero-value entries. The magnified details of the marked parts are shown in the bottom right of the pattern.

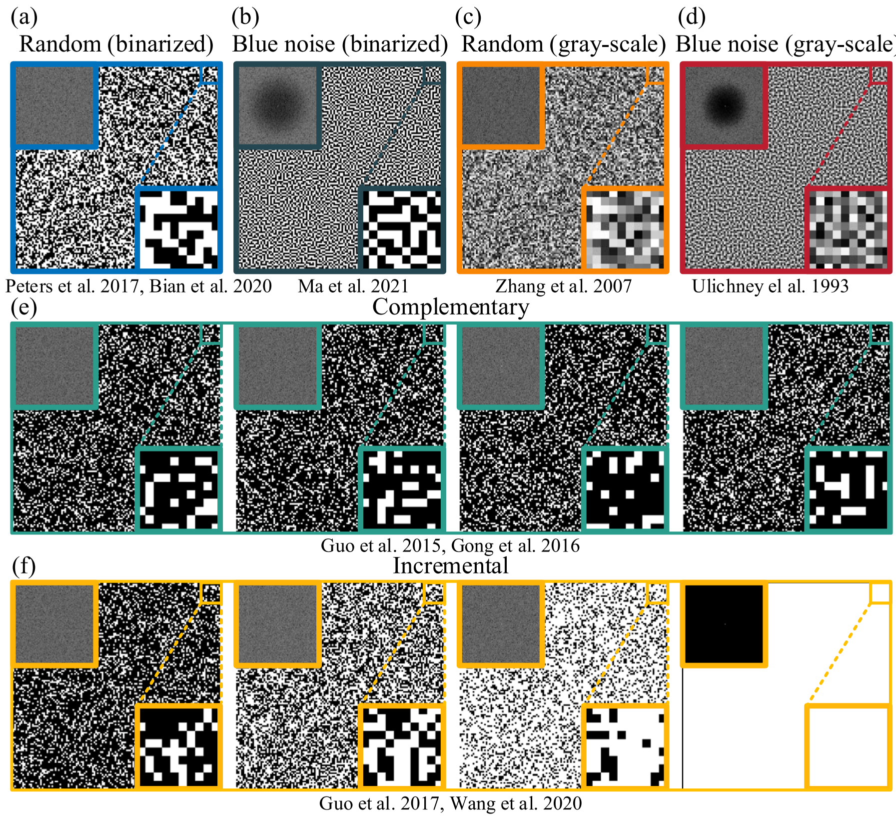

Fig. 2. Examples of (a) binarized random patterns, (b) binarized blue noise patterns, (c) gray-scale random patterns, (d) gray-scale blue noise patterns, (e) a group of complementary patterns with $K = 4$, and (f) a group of incremental patterns with $K = 4$.

For the case of amplitude modulation, each entry of P_k independently takes the value 0 or 1 with an equal probability of $\frac{1}{2}$. The inset map in the upper left shows the absolute value of the Fourier transform of the pattern on a logarithmical scale. Energy is spread out uniformly across all frequency bands. This means that large-scale structures of binarized random patterns are just as strong as small-scale structures.

For the phase modulation case, $P_k(x, y)$ is sampled from $\left\{e^{j0}, e^{j\Phi}\right\}$ with equal probability. Here (x, y) represents the discrete pixel coordinates, and Φ represents phase modulation depth, which can be any non-zero value. In the following sections, we will further investigate the optimal phase modulation depth.

2.2.2. Binarized blue noise patterns

It can be seen from Fig. 2(a) that the binarized random patterns tend to form clusters of ones and zeros, which means that the area includes only blocked or unblocked elements. Based on the concept of digital halftoning [51], the blue noise patterns try to spread all the blocked and unblocked elements as homogeneously as possible, as shown in Fig. 2(b). The blue noise texture has less large-scale structure than the binarized random patterns, and indeed the Fourier transform has low energy near the center, exhibiting spatiotemporal characteristics with high-frequency components and with little or no low-frequency components. Thus, the clusters disappear, and uniform sensing is achieved. In this work, we adopted a method based on the direct binary search (DBS) algorithm [52] to generate the binarized blue noise patterns. The algorithm was initialized with a constant tone of 0.5, and then entered an iterative process to attain the desired statistics. The method can ensure the spatial distribution of the blocked and unblocked elements to be aperiodic and homogeneous. At the same time, the generated pattern is very isotropic as can be seen from its radial symmetry around the center of the Fourier transform. Readers may refer to Ref. [53] for the operation details.

For the case of amplitude modulation, $P_k(x, y) \in \{0, 1\}$. For the phase modulation case, $P_k(x, y) \in \left\{e^{j0}, e^{j\Phi}\right\}$.

2.2.3. Gray-scale random patterns

Unlike binarized patterns that contain only two states, gray-scale random patterns are composed of uniformly distributed random numbers. Fig. 2(c) presents an exemplar gray-scale random pattern and its Fourier transform. For the case of amplitude modulation, P_k is an m-by-n matrix of uniformly distributed random numbers between 0 and 1. For the phase modulation case, $P_k(x, y)$ can be described as $e^{j\Phi\cdot\theta}$, where θ can be any number drawn from the uniform distribution in the interval $(0, 1)$.

2.2.4. Gray-scale blue noise patterns

As can be seen from Fig. 2(c), fairly obvious large-scale structures exist in the gray-scale random patterns. Instead, gray-scale blue noise patterns spread out the energy evenly in all directions and look almost entirely uniform, as shown in Fig. 2(d). In this work, the gray-scale blue noise patterns were pre-generated using the void and cluster method [54]. A SciPy implementation of the method and a precomputed database can be downloaded from the “Free blue noise textures” blog post (http://momentsingraphics.de/?p=127).

For the case of amplitude modulation, $P_k(x, y) \in (0, 1)$. For the phase modulation case, $P_k(x, y)$ can be described as $e^{j\Phi\cdot\theta}$, where $\theta \in (0, 1)$.

2.2.5. Complementary patterns

For the above four pattern structures, the random numbers are sampled from $\{0, 1\}$ with equal probability or $(0, 1)$ with uniform distribution, the transmission of the patterns is therefore $\frac{1}{2}$. We would also like to investigate illumination patterns with other transmittances. Here we consider complementary patterns [30,31], whose transmittance varies with the number of patterns. Assuming that the number of patterns is K, the transmittance of each pattern is accordingly $\frac{1}{K}$. For the case of amplitude modulation, the spatial distribution of each pattern is defined as

$$P_k(x, y) = \begin{cases} 1, for\ R(x, y) = k \\ 0, for\ R(x, y) \neq k \end{cases} \quad (k = 1, 2, \dots, K), \tag{2}$$

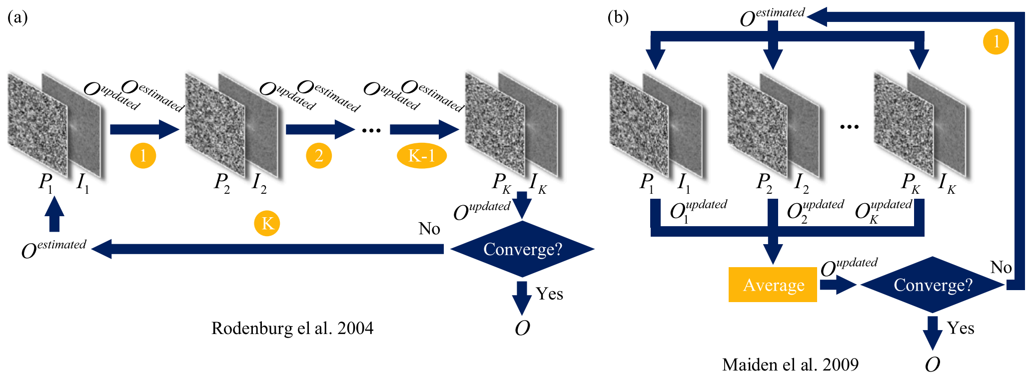

Fig. 3. Reconstruction framework of the algorithms. (a) The serial algorithm. (b) The parallel algorithm.

where $R(x, y)$ is a matrix with randomly distributed integers from 1 to K. Fig. 2(e) shows a group of patterns with $K = 4$. Evidently, such a group of patterns is complementary, which means that they have no overlapping pixels and satisfy

$$\sum_{k=1}^{K} P_k(x, y) = 1. \tag{3}$$

For the phase modulation case, the value distributions are described as

$$P_k(x, y) = \begin{cases} e^{j\Phi}, for\ R(x, y) = k \\ e^{j0}, for\ R(x, y) \neq k \end{cases} \quad (k = 1, 2, \ldots, K). \tag{4}$$

2.2.6. Incremental patterns

In Ref. [55], it was verified that complementary patterns tend to produce phase biases, which occur mainly at positions where the phase values change slowly. They further introduce the incremental constraint that is beneficial for eliminating the phase biases [32]. For the case of amplitude modulation, the spatial distribution of each pattern is defined as

$$P_k(x, y) = \begin{cases} 1, for\ R(x, y) \leq k \\ 0, for\ R(x, y) > k \end{cases} \quad (k = 1, 2, \ldots, K). \tag{5}$$

This modification allows that for a group of K illumination patterns, their transmittances are $\frac{1}{K}, \frac{2}{K}, \ldots, 1$, respectively, incremented by $\frac{1}{K}$ as a spacing. Fig. 2(f) presents an example of a group of incremental patterns with the number of patterns $K = 4$.

For the phase modulation case, the value distributions are described as

$$P_k(x, y) = \begin{cases} e^{j\Phi}, for\ R(x, y) \leq k \\ e^{j0}, for\ R(x, y) > k \end{cases} \quad (k = 1, 2, \ldots, K). \tag{6}$$

2.3. Reconstruction algorithms

With the knowledge of the illumination patterns P_k, the reconstruction aims to recover the complex-field object image O from the recorded intensity images I_k. As the field has advanced, numerous phase retrieval algorithms have been proposed [1,48], yet the early approaches have not seen major improvement and remain popular. The original PIE algorithm has been widely used in a range of applications, thanks to the benefits it offers in terms of easy coding, a small number of tuning parameters, and efficient memory use [56]. To investigate different types of illumination patterns in a unified framework, while avoiding the influence of factors such as parameter settings on the results, we adopt the PIE-based algorithms for reconstruction.

Deducted from the nonlinear algorithms of incremental gradient descent and global gradient descent, the PIE-based algorithms can be computed in a serial [21] or parallel [57] manner. The architectures of the serial and parallel algorithms are illustrated in Figs. 3(a) and (b), respectively. Also, the complete algorithms are summarized in Algorithms 1 and 2. The iterations begin with the initialization of the complex-field image. In our implementation, we used a matrix of ones as the initialized $O^{estimated}$ for a fair comparison. Then, the algorithms alternate back and forth through the object plane and detector plane to update the complex-field image O.

In each iteration, using the kth data pair of illumination pattern P_k and corresponding captured image I_k, the current complex-field image $O^{estimated}$ is multiplied with P_k, yielding the modulated object field $\psi_k = P_k \odot O^{estimated}$. Next, the modulated field is forward propagated to

Algorithm 1: Serial algorithm

Input: Captured images I_k, Illumination patterns P_k $(k = 1, 2, \ldots, K)$
Output: Complex-valued image O
Initialize $O^{estimated}$ with a matrix of ones;
while *not converge* **do**
  for $k = 1 : K$ **do**
    $\psi_k = P_k \odot O^{estimated}$;
    $\phi_k = \mathcal{F}(\psi_k)$;
    $\phi_k^{updated} = \sqrt{I_k} \cdot \phi_k / |\phi_k|$;
    $\psi_k^{updated} = \mathcal{F}^{-1}(\phi_k^{updated})$;
    $O^{updated} = O^{estimated} + P_k^* / |P_k|_{\max}^2 \cdot \left(\psi_k^{updated} - \psi_k\right)$;
    $O^{estimated} = O^{updated}$;
  end
end

Algorithm 2: Parallel algorithm

Input: Captured images I_k, Illumination patterns P_k $(k = 1, 2, \ldots, K)$
Output: Complex-valued image O
Initialize $O^{estimated}$ with a matrix of ones;
while *not converge* **do**
  for $k = 1 : K$ **do**
    $\psi_k = P_k \odot O^{estimated}$;
    $\phi_k = \mathcal{F}(\psi_k)$;
    $\phi_k^{updated} = \sqrt{I_k} \cdot \phi_k / |\phi_k|$;
    $\psi_k^{updated} = \mathcal{F}^{-1}(\phi_k^{updated})$;
    $O_k^{updated} = O^{estimated} + P_k^* / |P_k|_{\max}^2 \cdot \left(\psi_k^{updated} - \psi_k\right)$;
  end
  $O^{updated} = \frac{1}{K} \cdot \sum_{k=1}^{K} O_k^{updated}$;
  $O^{estimated} = O^{updated}$;
end

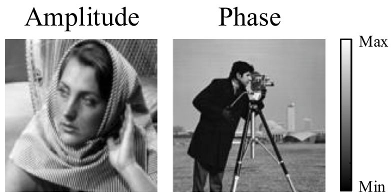

Fig. 4. Ground-truth images used as the object.

the detector plane using the Fourier transform to produce $\phi_k = \mathcal{F}(\psi_k)$. The captured image I_k is then used to implement modulus constraint to obtain the updated wavefront on the detector plane, which is $\phi_k^{updated} = \sqrt{I_k} \cdot \phi_k / |\phi_k|$. After that, the wavefront is propagated back to the object plane and yields $\psi_k^{updated} = \mathcal{F}^{-1}(\phi_k^{updated})$. The object field is then demodulated using a formula from the PIE algorithm, and a refined estimation of O is obtained, as shown in line 8 of Algorithms 1 and 2. The above process is repeated for all K data pairs to complete one iteration.

In each iteration of the serial algorithm, the complex-field image O is sequentially updated by K pairs of illumination patterns $\{P_1, P_2, \dots, P_K\}$ and corresponding captured images $\{I_1, I_2, \dots, I_K\}$. The updating procedure of O performs K times in one iteration. This ensures fast convergence of the serial algorithm. However, since only one pair of data is used for each update, the serial algorithm is susceptible to noise.

In the parallel algorithm, the average of the refined results $\{O_1^{updated}, O_2^{updated}, \dots, O_K^{updated}\}$ separately obtained by all K data pairs is used to update O. This averaging operation helps to suppress noise and enhance robustness. However, the convergence speed of the parallel algorithm is reduced since the complex-field image O is updated only once in each iteration.

3. Results

In this section, we compare the performance of the above types of illumination patterns—from the aspects of imaging quality, imaging efficiency, and robustness to noise, using both serial and parallel algorithms. The cases of noiseless, additive noise (quantization noise and Gaussian noise), and multiplicative noise (Poisson noise and speckle noise) are considered. In the following numerical simulations, the amplitude and phase of the ground truth were generally taken as two standard images, "Barbara" (normalized to $[0,1]$) and "Cameraman" (normalized to $[0,2\pi]$), respectively, as shown in Fig. 4. The image size was set as 128 × 128 pixels. For each test case, we repeated the simulations 50 times and averaged the evaluations to produce final quantitative results. For a fair comparison, the iteration stopped when the mean square error (MSE) of two successive reconstructions was smaller than 10^{-18}, or the maximum iteration number 5000 was reached. All the simulations were implemented using MATLAB 2016a with a 3.6 GHz Intel Core i7 processor and 16 GBRAM.

3.1. Noiseless cases

In most situations, a larger number of modulations should be better for more accurately reconstructing the target wavefront. In real experiments, however, we always hope for the least pattern number to achieve the required accuracy. To explore the possible optimal range of the number of patterns, we first investigate the imaging quality of different types of illumination patterns with respect to the number of modulations ranging from 2 to 20. The noiseless diffraction patterns were synthesized according to Eq. (1). Both the serial and parallel algorithms were used for reconstruction. The structural similarity index measure (SSIM) [58] was used to evaluate the reconstruction quality quantitatively, by calculating that between the reconstructed amplitude and its ground truth. The SSIM index ranges from 0 to 1, where 1 corresponds to identical images and 0 represents a loss of all similarity.

3.1.1. Amplitude modulation

The quantitative reconstruction results are shown in Fig. 5. Detailed comparisons and analysis are available in Sec. S1.1 of the Supplementary Material. According to the results, we conclude that without measurement noise, imaging efficiency of the binarized random and blue noise patterns is the highest, enabling fast and lossless reconstruction under only 3 projections. The complementary and incremental patterns achieve nearly lossless reconstructions with relatively small pattern numbers (4 and 3, respectively). However, the reconstruction quality is not satisfactory in the case where the number of patterns is large. The imaging efficiency of the gray-scale patterns is lower than that of the binarized ones. Six patterns are required to obtain nearly lossless reconstructions, while a dozen of patterns is required to achieve lossless reconstructions. Except for the case where complementary patterns are used, the parallel algorithm provides higher reconstruction accuracy than the serial algorithm, but at the cost of worse convergence performance.

3.1.2. Phase modulation

Different from amplitude modulation, phase modulation involves modulation depth, which represents the range of phase variation in the illumination patterns. In this subsection, we first compared the six types of patterns under different phase modulation depths to find the optimal modulation depths for the following experiments. The quantitative results are presented in Fig. 6. Detailed comparisons and analysis are available in Sec. S1.2 of the Supplementary Material. From the simulations, it can be concluded that for binarized phase modulation, the optimal modulation depth is π. For gray-scale phase modulation, the optimal modulation depth is 2π. With the optimal phase modulation depth, the serial algorithm yields lossless reconstructions using only 3 patterns of any type. However, the parallel algorithm guarantees fast convergence only when the number of patterns is no less than 4. The blue noise patterns (both binarized and gray-scale) and incremental patterns are advantageous in the case where the pattern number is 3. We can further conclude that these types of patterns facilitate algorithmic convergence.

3.2. Quantization noise cases

In most diffraction imaging systems, a charge coupled device (CCD) or complementary metal oxide semiconductor (CMOS) is used to record the diffraction images. The digital detectors convert the analog signal to a digital one with n bits of accuracy [59]. Here n is the bit depth of the sensor, which represents the number of shades of gray that can be recorded. The standard sensors have 8-bit depth, providing a quantified intensity between 0 and 255. Some advanced sensors can record images in 18-bit depth. Therefore, different levels of quantization noise exist in practical applications due to analog-to-digital conversion.

To compare the performance of different types of illumination patterns in terms of robustness to quantization noise, we artificially quantized the diffraction images to be captured, to bits ranging from 8 to 18. Specifically, we scaled the noiseless diffraction images by $2^n - 1$ and rounded them to the nearest integer values, which is mathematically described as:

$$I_k^{quantization} = \left[\frac{\left|\mathcal{F}\left(P_k \odot O\right)\right|^2}{\left|\mathcal{F}\left(P_k \odot O\right)\right|^2_{\max}} \cdot (2^n - 1) \right], \tag{7}$$

where $[\cdot]$ represents the rounding operation.

According to the results in Section 3.1, we compared the different types of patterns in three cases where the pattern numbers are 3, 6, and 18. Specifically, 3 is the minimum number of patterns required for reconstruction, 6 is the least number to achieve successful reconstructions for all the patterns, and 18 patterns ensure high-quality reconstruction using all types of patterns. Both the peak signal-to-noise ratio (PSNR) and SSIM index were used as quantitative evaluation metrics for reconstruction quality.

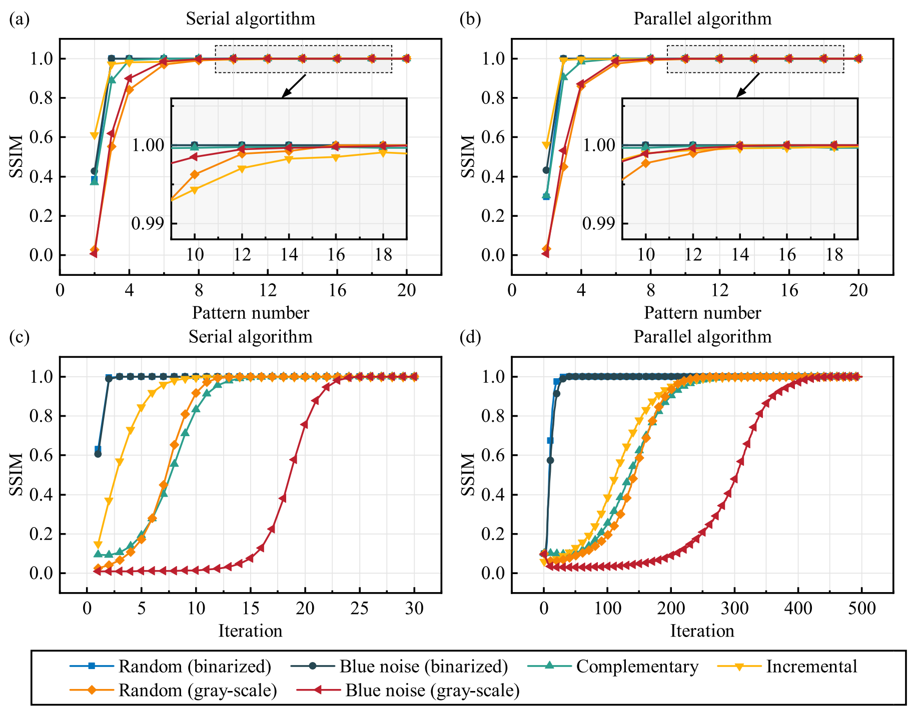

Fig. 5. Quantitative comparison of amplitude patterns in the noiseless cases. (a) SSIM index versus pattern number using the serial algorithm. (b) SSIM index versus pattern number using the parallel algorithm. (c) SSIM index versus iteration when the pattern number is 18 using the serial algorithm. (d) SSIM index versus iteration when the pattern number is 18 using the parallel algorithm.

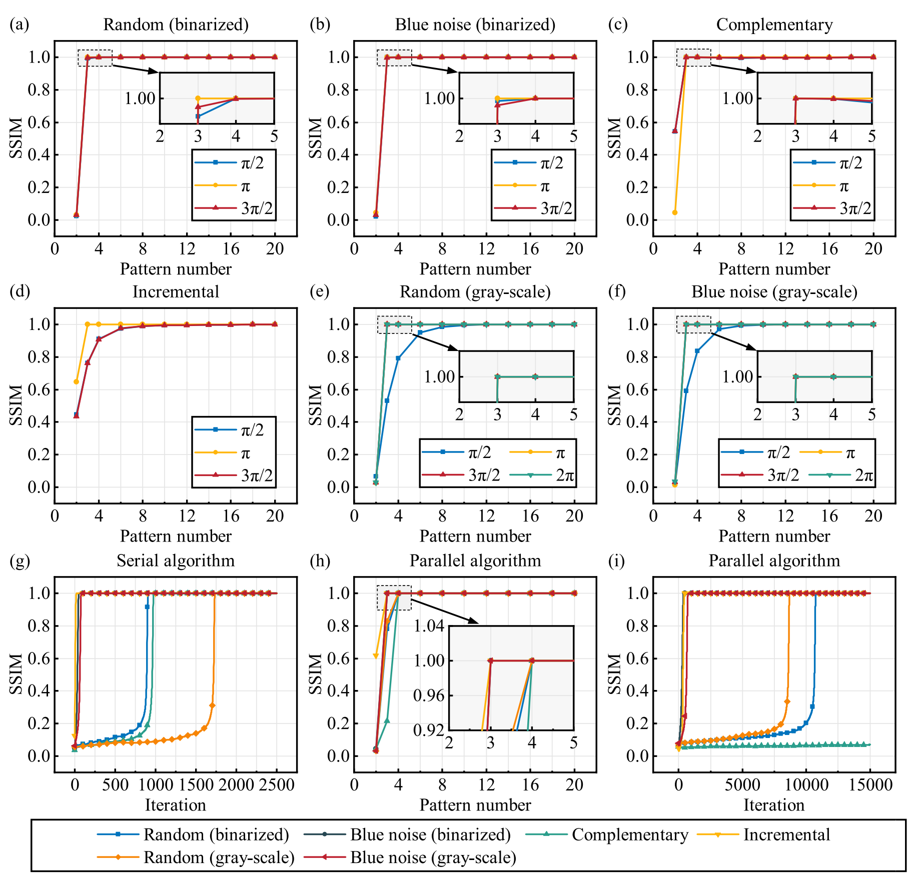

Fig. 6. Quantitative comparison of phase patterns in the noiseless cases. (a)–(f) SSIM index versus number of (a) binarized random, (b) binarized blue noise, (c) complementary, (d) incremental, (e) gray-scale random, and (f) gray-scale blue noise patterns using the serial algorithm. (g) SSIM index versus iteration when the pattern number is 3 using the serial algorithm. (h) SSIM index versus pattern number using the parallel algorithm. (i) SSIM index versus iteration when the pattern number is 3 using the parallel algorithm.

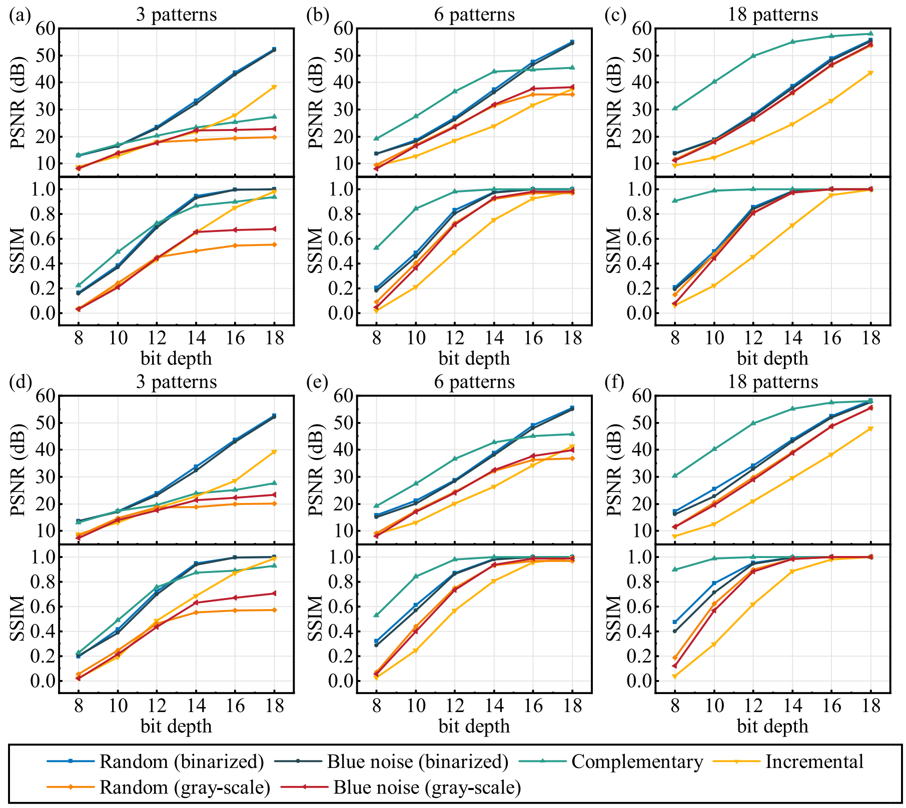

Fig. 7. Quantitative comparison of amplitude patterns under quantization noise. (a)–(c) PSNR and SSIM index versus bit depth using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus bit depth using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

3.2.1. Amplitude modulation

The quantitative reconstruction results are shown in Fig. 7. Qualitative visualization results and detailed analysis are available in Sec. S2.1 of the Supplementary Material. According to the results, we conclude that coded CDI based on amplitude modulation is generally sensitive to quantization noise. The use of the parallel algorithm helps slightly enhance its noise robustness. In high-speed imaging applications, to obtain satisfactory results, 3 binarized random or blue noise patterns and a sensor with a bit depth of no less than 16 can be used for imaging. In applications where the dynamic range of the sensor is limited, the transmittance of the binarized patterns can be reduced appropriately to obtain stronger robustness to quantization noise. Besides, incremental patterns should be avoided in applications where the bit depth of the sensor is low.

3.2.2. Phase modulation

The quantitative reconstruction results are shown in Fig. 8. Qualitative visualization results and detailed analysis are available in Sec. S2.2 of the Supplementary Material. According to the results, it can be concluded that, when using the optimal phase modulation depths, coded CDI based on phase modulation using pattern structures with a transmittance of $\frac{1}{2}$ is generally robust to quantization noise. In high-speed imaging applications, the random and blue noise patterns (both binarized and gray-scale) and the serial algorithm are the best choices. In high-accuracy imaging applications, a large number of phase patterns and the parallel algorithm can be considered. Besides, complementary and incremental patterns should be avoided when the bit depth of the sensor is limited.

3.3. Gaussian noise cases

The additive white Gaussian noise is the most prevalent noise in imaging. It occurs during acquisition and transmission through analog circuitry, which is pretty much the basic operation required for computerization [60]. Gaussian noise is a part of almost any signal, its noise intensity is uniform and is independent of the ground truth signal. In this subsection, the resulting measurements are assumed to be contaminated by white Gaussian noise, which follows the probability distribution

$$P(N_{gaussian}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{N_{gaussian}^2}{2\sigma^2}\right), \tag{8}$$

where $N_{gaussian}$ stands for noise, and σ^2 is its variance.

To compare the performance of different types of illumination patterns in terms of robustness to Gaussian noise, white Gaussian noise was added to the diffraction images with a variance ranging from 10^{-12} to 10^{-4}. Specifically, the range of the diffraction images was first linearly mapped into $[0, 1]$, and then Gaussian noise in different levels was added to the diffraction images which then were linearly mapped back to the same range as the measurements.

3.3.1. Amplitude modulation

The quantitative reconstruction results are shown in Fig. 9. Qualitative visualization results and detailed comparative analysis are available in Sec. S3.1 of the Supplementary Material. In simple terms, coded CDI based on amplitude modulation is generally susceptible to Gaussian noise. Binarized random and blue noise patterns outperform the others when the least number of patterns is used. This is advantageous in high-speed imaging applications. The complementary

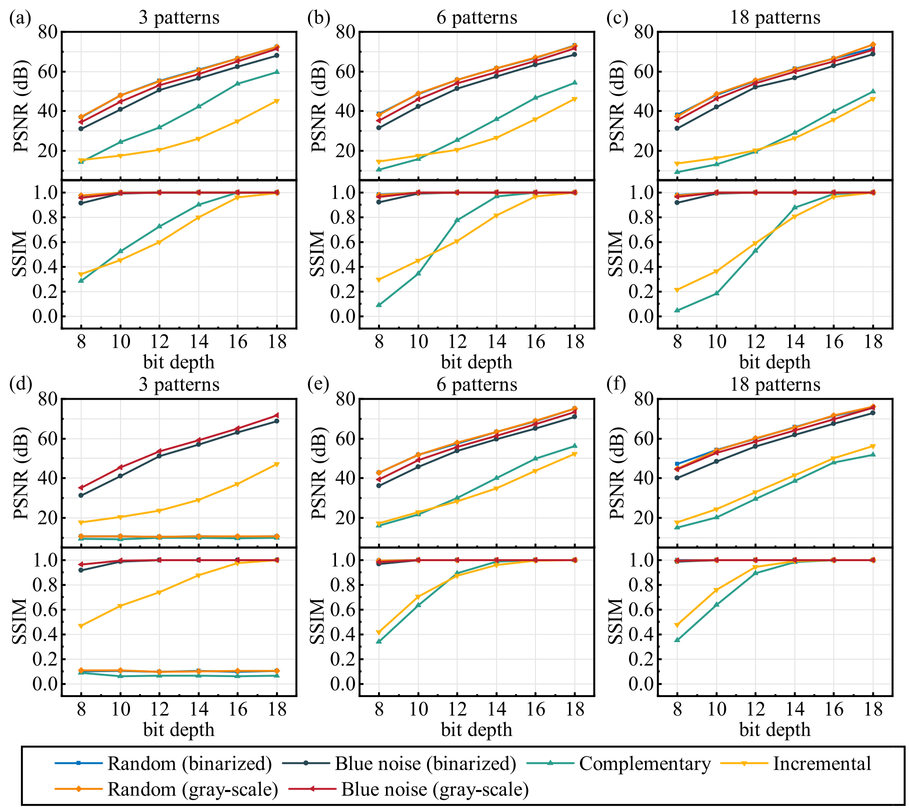

Fig. 8. Quantitative comparison of phase patterns under quantization noise. (a)–(c) PSNR and SSIM index versus bit depth using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus bit depth using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

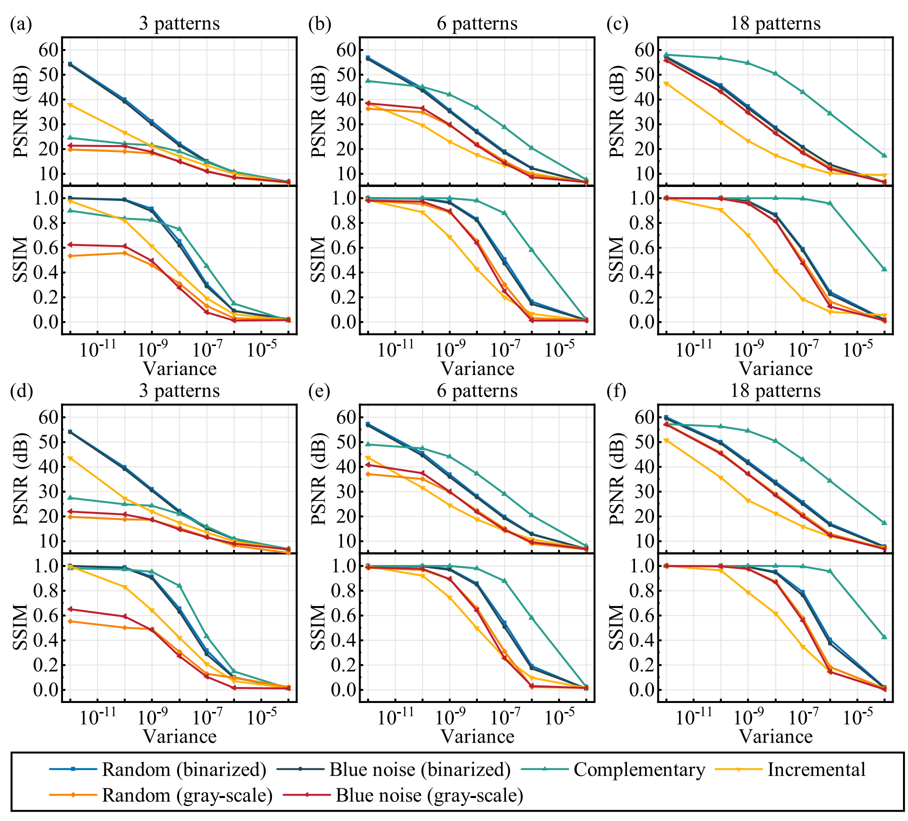

Fig. 9. Quantitative comparison of amplitude patterns under Gaussian noise. (a)–(c) PSNR and SSIM index versus noise variance using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus noise variance using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

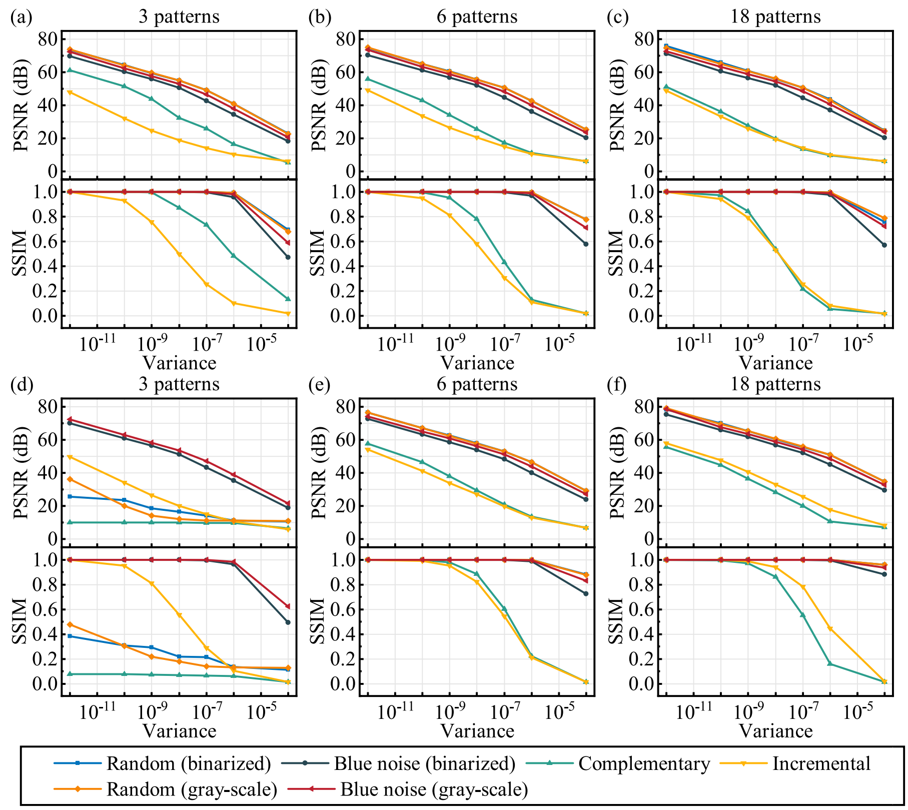

Fig. 10. Quantitative comparison of phase patterns under Gaussian noise. (a)–(c) PSNR and SSIM index versus noise variance using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus noise variance using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

patterns give the best reconstruction when using a sufficient number of projections, especially under heavy Gaussian noise. Therefore, in practical applications with strong Gaussian noise, one can appropriately reduce the transmittance of the binarized amplitude patterns to weaken the zero-order diffraction, thereby obtaining stronger noise robustness.

3.3.2. Phase modulation

The quantitative results are shown in Fig. 10. Qualitative visualization results and detailed analysis are available in Sec. S3.2 of the Supplementary Material. In short, considering complementary and incremental patterns remarkably underperforms the others under Gaussian noise, one should avoid using them in such cases. When using the optimal phase modulation depths and the remaining four types of patterns, coded CDI based on phase modulation is generally robust to Gaussian noise with noise variance below 10^{-6}. The random patterns are advantageous under heavy Gaussian noise with noise variance above 10^{-6}. The blue noise patterns facilitate the convergence of the parallel algorithm when the pattern number is small.

3.4. Poisson noise cases

Image sensors measure scene irradiance by counting the number of discrete photons incident on the sensor over a given time interval. Poisson noise, also termed Photon noise, is a basic form of uncertainty associated with the independence of random individual photon arrivals [61]. The number of photons S measured by a given sensor element over a time interval t is described by the discrete probability distribution

$$P(S = l) = \frac{e^{-\lambda t}(\lambda t)^l}{l!}, \tag{9}$$

where λ is the expected number of photons per unit time interval, which is proportional to the incident scene irradiance. Poisson noise is considered as a signal-dependent multiplicative noise and has a variance that scales linearly with the signal magnitude. Even under ideal imaging conditions, free from all other sensor-based sources of noise, any measurement would still be subject to Poisson noise.

Poisson noise is inherent to the measurement of light, and has no parameters to be calibrated. As a result, the effect of Poisson noise on imaging can be characterized using the radiometric response function that relates the photon count and the expected pixel intensity. To compare the performance of different types of illumination patterns in terms of robustness to Poisson noise, Poisson noise was added to the diffraction images with a flux ranging from 10^4 to 10^9 photons. Specifically, we first linearly mapped the range of the diffraction images so that the sum of all pixel intensities was the given flux. Then Poisson noise for different fluxes was added to the diffraction images which then were linearly mapped back to the same range as the measurements.

3.4.1. Amplitude modulation

The quantitative reconstruction results are shown in Fig. 11. Qualitative visualization results and detailed comparative analysis are available in Sec. S4.1 of the Supplementary Material. To summarize, Poisson noise is inherent to the measurement of light, and imaging quality improves with increasing flux. Under Poisson noise, the difference in the performance of the patterns is noticeable when the pattern number is small, and becomes insignificant when the pattern number is large. In cases where the number of patterns is small, binarized random and blue noise patterns are good choices. As for cases where the pattern number is large, except for the incremental patterns, little difference is found between different types of patterns. In high-precision imaging applications, the parallel algorithm is a suitable reconstruction strategy. Besides, incremental patterns should be avoided under low-light conditions.

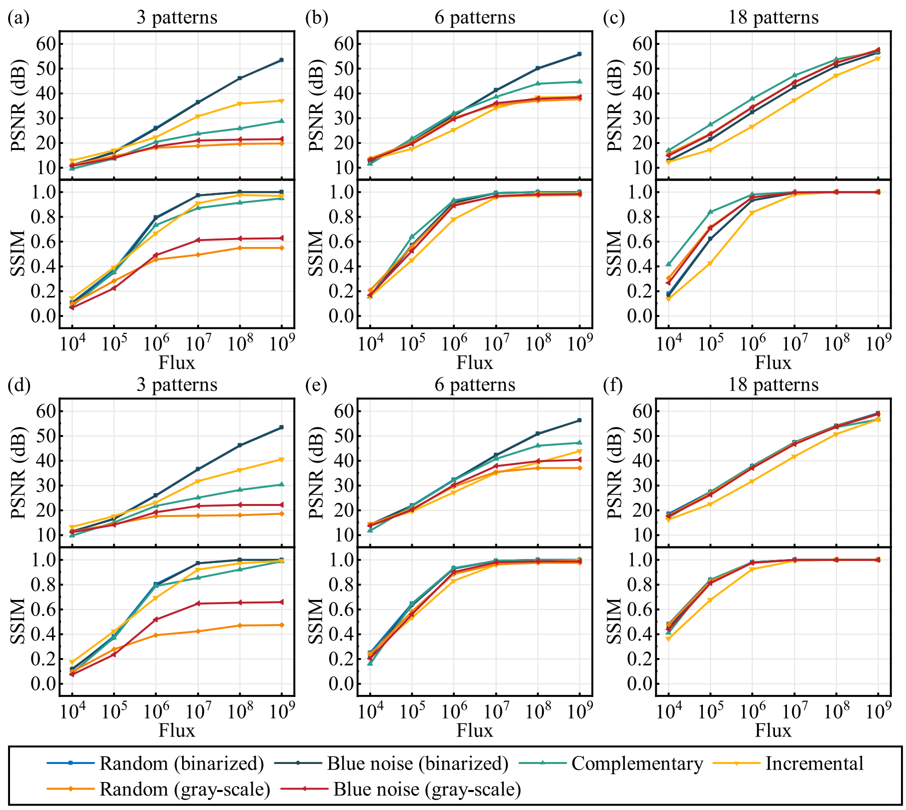

Fig. 11. Quantitative comparison of amplitude patterns under Poisson noise. (a)–(c) PSNR and SSIM index versus flux using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus flux using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

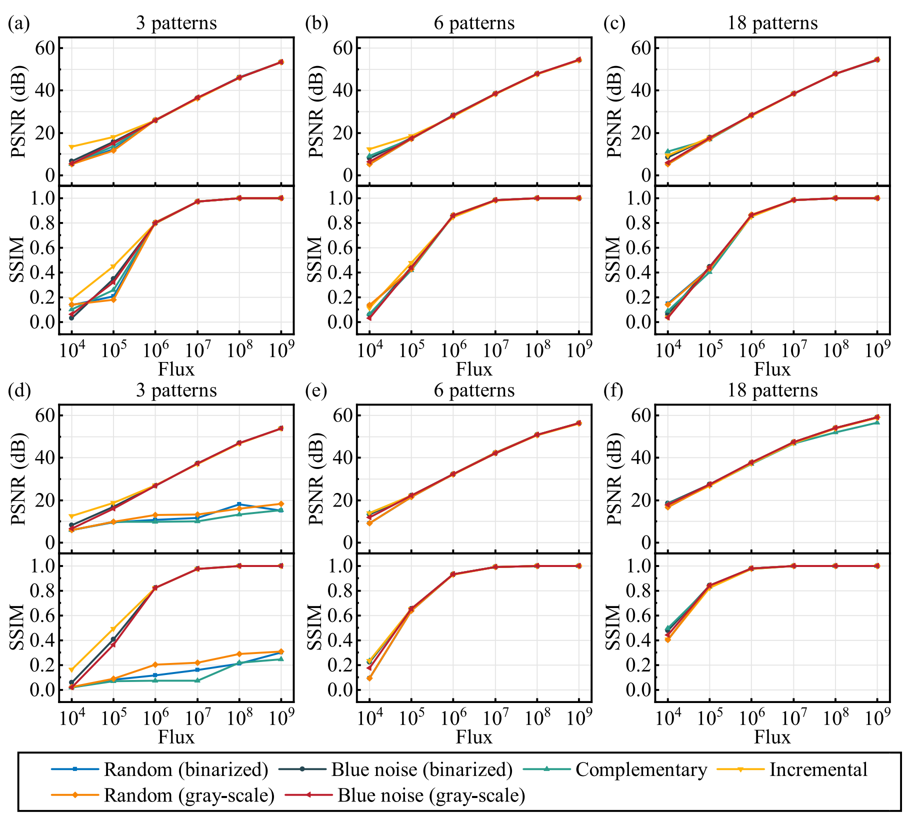

Fig. 12. Quantitative comparison of phase patterns under Poisson noise. (a)–(c) PSNR and SSIM index versus flux using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus flux using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

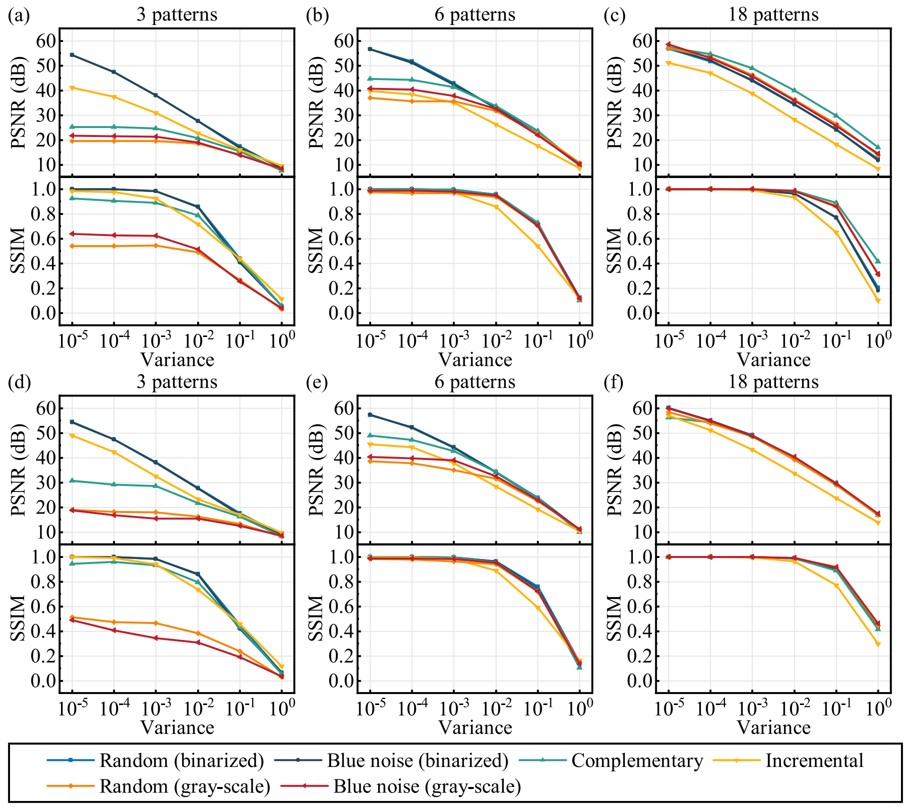

Fig. 13. Quantitative comparison of amplitude patterns under speckle noise. (a)–(c) PSNR and SSIM index versus noise variance using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus noise variance using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

3.4.2. Phase modulation

The quantitative results are shown in Fig. 12. Qualitative visualization results and detailed comparative analysis are available in Sec. S4.2 of the Supplementary Material. To summarize, for coded CDI based on phase modulation, in terms of robustness to Poisson noise, the difference between different types of patterns is in general not significant. In cases where the number of patterns is small, the incremental patterns show a slight advantage under low fluxes. In other cases, the performance of all types of phase patterns is highly consistent.

3.5. Speckle noise cases

Speckle is an inherent characteristic of images acquired with imaging techniques based on the detection of coherent waves. Speckle noise is a fluctuation of intensity over an image caused by very irregular spatial interference from the coherent phases [62]. Since our coded CDI system is based on coherent imaging, it is inherently associated with speckle noise.

Speckle noise is a type of multiplicative noise. Its probability density function follows a gamma distribution. Practical implementation of adding the speckle noise can be performed using the equation

$$I^{speckle} = I + N_{uniform} \odot I, \tag{10}$$

where $N_{uniform}$ is uniformly distributed random noise with mean 0 and a preset variance [63].

To compare the performance of different types of illumination patterns in terms of robustness to speckle noise, speckle noise was added to the diffraction images with a variance ranging from 10^{-5} to 10^{0}. Specifically, the range of the diffraction images was first linearly mapped into $[0, 1]$, and then speckle noise in different levels was added to the diffraction images which then were linearly mapped back to the same range as the measurements.

3.5.1. Amplitude modulation

The quantitative reconstruction results are shown in Fig. 13. Qualitative visualization results and detailed comparative analysis are available in Sec. S5.1 of the Supplementary Material. Simply put, the difference in robustness to speckle noise for different types of patterns is noticeable when the pattern number is small, and becomes insignificant when the pattern number is large. Binarized random and blue noise patterns outperform the others when using the least number of patterns. This is advantageous in high-speed imaging applications. In cases where the number of patterns is large, except for the incremental patterns, little difference is found between different types of patterns. The parallel algorithm is more robust to speckle noise than the serial algorithm, and further narrows the difference between the patterns. Besides, one should avoid using incremental patterns under heavy speckle noise.

3.5.2. Phase modulation

The quantitative results are shown in Fig. 14. Qualitative visualization results and detailed comparative analysis are available in Sec. S5.2 of the Supplementary Material. In short, for coded CDI based on phase modulation, in terms of robustness to speckle noise, the difference between the patterns is in general not significant. Incremental patterns show a slight advantage under heavy noise when the least pattern number is used. In other cases, the performance of all types of phase patterns is highly consistent.

4. Discussion and conclusions

In this study, six representative illumination pattern structures for coded CDI are compared in terms of imaging quality, imaging efficiency, and robustness to noise, using both serial and parallel reconstruction algorithms. According to the results, several conclusions can be drawn.

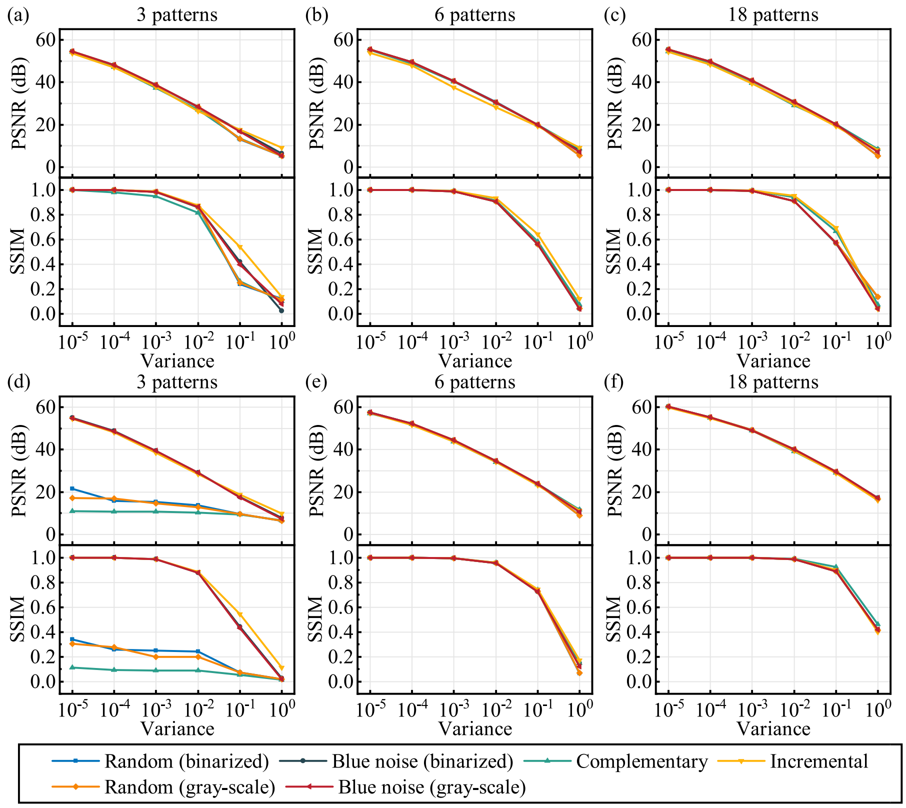

Fig. 14. Quantitative comparison of phase patterns under speckle noise. (a)–(c) PSNR and SSIM index versus noise variance using (a) 3 patterns, (b) 6 patterns, and (c) 18 patterns for illumination and the serial algorithm for reconstruction. (d)–(f) PSNR and SSIM index versus noise variance using (d) 3 patterns, (e) 6 patterns, and (f) 18 patterns for illumination and the parallel algorithm for reconstruction.

(1) In noiseless cases, using amplitude modulation, the reconstruction quality is mainly dependent on the number of illumination patterns. To obtain successful reconstructions (the average PSNR of 50 simulations higher than 30 dB, SSIM index above 0.9), the least number of patterns required is 3. To obtain high-quality reconstructions using all types of patterns, the number of patterns required is 18. Using phase modulation, the reconstruction quality is further influenced by the modulation depth. For higher reconstruction quality using the least pattern number, the optimal phase modulation depth is π in the case of binarized phase modulation and 2π in the case of gray-scale phase modulation.
(2) The parallel algorithm in general provides higher reconstruction quality than the serial one, but at the cost of worse convergence performance. One exception is phase modulation using 3 patterns, where the parallel algorithm cannot guarantee convergence within the maximum number of iterations, while the serial algorithm enables convergence, hence the serial algorithm gives higher reconstruction quality. The convergence performance of the parallel algorithm is highly relevant to the type of phase patterns. The blue noise and incremental phase patterns facilitate the algorithmic convergence.
(3) Considering capture efficiency, binarized random and binarized blue noise amplitude patterns are the best choices, allowing for promising reconstruction quality under only 3 projections. The binarized amplitude modulation can be implemented at a refreshing rate of up to 30 kHz using a DMD.
(4) Considering zero-mean additive noise (quantization noise and Gaussian noise), patterns that can produce diffraction images without strong zero order outperform the others. Under heavy noise, the preferred amplitude patterns are complementary patterns with low transmittance, and the advantageous phase patterns are the random patterns.
(5) Considering multiplicative noise (Poisson noise and speckle noise), the difference between different types of patterns is insignificant when using a large number of patterns. Under amplitude modulation conditions, the incremental patterns underperform the others, while little difference is found between the others. Under phase modulation conditions, the performance of the patterns is highly consistent.

According to the principal analysis and results, each scheme or algorithm has its own advantages and limitations. Their characteristics can be further combined to improve imaging quality. For instance, we find that complementary amplitude patterns are advantageous under heavy additive noise. However, phase biases occur due to the absence of overlapping area constraints. To circumvent this limitation, we can divide the patterns into several sub-groups and constrain the patterns in every sub-group to be complementary. The patterns belonging to different sub-groups are not necessarily complementary. In this way, the low transmittance and overlapping area constraints are satisfied simultaneously. Additionally, blue noise and incremental phase patterns contribute to the convergence of the parallel algorithm, while random phase patterns are beneficial under heavy additive noise. In high-precision imaging applications, a mixture of random and blue noise phase patterns can be used, and the parallel algorithm can be adopted, to balance imaging quality and computation time. In a word, one can choose appropriate illumination patterns and reconstruction algorithms according to specific coded CDI configurations and applications.

Finally, we would like to note that there are three future directions for the advancement of our present effort. (1) In this work, six regular pattern structures were studied. More pattern structures such as optimized patterns [64], speckle patterns [65], and green noise patterns [66] can be further investigated for higher imaging accuracy. (2) The effects of experimental parameters such as the exposure time of the camera, the sampling effect of the modulator, and the numerical aperture of the microscope objective can be further considered. (3) The PIE-based algorithms were adopted in this work, for the convenience of comparing different types of patterns in a unified framework. Similarly, for the selected illumination patterns, the performance of more phase retrieval algorithms [1,67] can be further compared to choose the appropriate algorithm for reconstruction.

CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Tong Qin:** Writing – review & editing, Funding acquisition. **Zhijie Gao:** Supervision, Writing – review & editing, Funding acquisition. **Liheng Bian:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.optlastec.2023.109861.

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